Mersenne Exponents: DR Patterns vs the 7-4-9 Triad

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Rebuilt from the logged title after the original was lost; the original wording was not recoverable from past chats, so every definition below is stated explicitly.

1. Where the 7-4-9 triad sits

The digital root of $2^n - 1$ repeats every 6 steps, because 2 has order 6 mod 9:

n mod 6	1	2	3	4	5	0
DR(2^n)	2	4	8	7	5	1
DR($2^n - 1$)	1	3	**7**	6	**4**	**9**

Definition (the 7-4-9 triad). The roots 7, 4 and 9 that $2^n - 1$ takes at $n \equiv 3, 5$ and 0 (mod 6): the second half of the cycle 1, 3, 7, 6, 4, 9 with the 6 removed.

The cycle splits cleanly by parity of n: odd n gives 1, 7, 4 (all $\equiv 1 \bmod 3$), even n gives 3, 6, 9 (all $\equiv 0 \bmod 3$). Every prime exponent except 2 is odd, so Mersenne numbers $M_p = 2^p - 1$ can only land on 1, 7 or 4 once $p > 2$.

2. Theorem: the exponent's family fixes DR(M_p)

Theorem. For every prime $p \geq 5$:

- if DR(p) $\in \{1, 4, 7\}$ ($p \equiv 1 \bmod 3$), then DR(M_p) = **1**;
- if DR(p) $\in \{2, 5, 8\}$ ($p \equiv 2 \bmod 3$), then DR(M_p) = **4**.

The two small cases are p = 2 (DR(M_2) = 3) and p = 3 (DR(M_3) = 7).

Proof. A prime $p \geq 5$ is odd and not divisible by 3, so $p \equiv 1$ or 5 (mod 6), and $p \equiv 1 \pmod 6$ exactly when $p \equiv 1 \pmod 3$. By the cycle in Section 1, $n \equiv 1$ gives DR($2^n - 1$) = 1 and $n \equiv 5$ gives 4. Since $9 \equiv 0 \pmod 3$, DR(p) $\in \{1, 4, 7\}$ is the same as $p \equiv 1 \pmod 3$. ∎

p mod 18	DR(p)	p mod 6	DR(M_p)
1	1	1	1
7	7	1	1

p mod 18	DR(p)	p mod 6	DR(M_p)
13	4	1	1
5	5	5	4
11	2	5	4
17	8	5	4

Corollaries against the triad.

1. **9 is impossible.** DR($2^n - 1$) = 9 requires 6 | n, which no prime satisfies.
2. **7 occurs exactly once,** at p = 3, the only prime ≡ 3 (mod 6).
3. **4 is the live member of the triad:** it is hit by every exponent in the 2-5-8 family, and only by them.
4. The 1-4-7 exponent family maps entirely outside the triad, to root 1.

3. All 52 known exponents against the triad

Of the 52 known Mersenne primes, 27 land on a triad root (26 on 4, 1 on 7, 0 on 9) and 25 land outside it (24 on 1, 1 on 3).

DR(M_p)	In triad?	Count	Exponents giving it
4	yes	26	every exponent with DR(p) ∈ {2, 5, 8} except p = 2
7	yes	1	p = 3
9	yes	0	none possible
1	no	24	every exponent with DR(p) ∈ {1, 4, 7}
3	no	1	p = 2

The exponent digital roots, split by family:

Family	DR(p) counts	Total	Maps to
1-4-7	1: 11, 4: 8, 7: 5	24	DR(M_p) = 1
2-5-8	2: 9, 5: 12, 8: 6	27 (26 for p ⩾ 5)	DR(M_p) = 4
3-6-9	3: 1, 6: 0, 9: 0	1	DR(M_p) = 7

DR(p) = 5 is the single largest class (12), and DR(p) = 6 and 9 are empty because no prime above 3 is divisible by 3.

4. Is there a bias?

No statistically detectable bias: the 52 exponents split between the families and across the six possible roots about as ordinary primes do.

Baseline. Among all primes from 5 to 10,000,000, the six roots 1, 2, 4, 5, 7, 8 each hold 16.65% to 16.68%, and 49.99% of primes are ≡ 1 (mod 3).

Test (exponents p ⩾ 5, n = 50)	Observed	Expected	Result
1-4-7 vs 2-5-8 family	24 vs 26	25 vs 25	two-sided binomial p = 0.89
Six roots vs uniform	11, 8, 8, 12, 5, 6	8.33 each	$\chi^2 = 4.48$ on 5 df, p = 0.48

So the triad split is a direct consequence of the theorem (each exponent's family decides its root), and the family sizes themselves look like a fair draw. The DR = 5 lead (12 of 50) is within ordinary variation for a sample this size; 50 exponents cannot separate a real tilt of a few percent from noise.

5. Ties to the 3-6-9 build and the mod-37 field

DR(p) alone does determine DR(M_p) for p ⩾ 5. DR(p) fixes p mod 9, hence p mod 3; an odd prime's class mod 6 is fixed by its class mod 3. Primes sharing a digital root cannot fall in different classes mod 6 (checked: zero exceptions among all primes 5 to 100,000). The joint table in Section 2 is the full map.

Same clock as the 3-6-9 build.

- The 6-cycle 1, 3, 7, 6, 4, 9 of DR($2^n - 1$) is the doubling cycle 2, 4, 8, 7, 5, 1 (Lemma 6 of the build) shifted down by one.
- Odd n gives the 1-7-4 family, even n gives the 3-6-9 family. Prime exponents are odd, so like the insertion rule's roots (Theorem 3d of the build) the Mersenne roots never enter 3, 6 or 9 after the first two primes; 9 at n ≡ 0 is the "nine is zero" state (P3) that a prime exponent can never reach.
- Exponent root lives mod 9 and Mersenne root mod 6; together they live mod 18. That is the same 18 that runs the insertion rule's root cycle, for the same reason: 2 has order 6 mod 9 and order 18 mod 27.

Mod-37 field. 2 is a primitive root mod 37 (order 36), so M_p mod 37 is fixed by p mod 36, and 37 divides M_p only when 36 divides p, which no prime does. Since p mod 36 is the pair (p mod 4, p mod 9), one residue class carries DR(p), DR(M_p) and M_p mod 37 together. The 52 exponents fill all 12 prime classes mod 36 and give 14 distinct values of M_p mod 37:

M_p mod 37	1	3	4	7	12	14	16	17	18	19	21	23	31	34
Exponents	4	1	3	1	1	4	2	5	1	3	4	7	9	7

6. The 52 exponents

Every row obeys the theorem: p mod 6 = 1 gives DR(M_p) = 1, and p mod 6 = 5 gives DR(M_p) = 4.

#	p	DR(p)	p mod 6	DR(M_p)	In triad
1	2	2	2	3	no
2	3	3	3	7	yes
3	5	5	5	4	yes
4	7	7	1	1	no
5	13	4	1	1	no
6	17	8	5	4	yes
7	19	1	1	1	no
8	31	4	1	1	no
9	61	7	1	1	no
10	89	8	5	4	yes
11	107	8	5	4	yes
12	127	1	1	1	no
13	521	8	5	4	yes
14	607	4	1	1	no
15	1,279	1	1	1	no
16	2,203	7	1	1	no
17	2,281	4	1	1	no
18	3,217	4	1	1	no
19	4,253	5	5	4	yes

#	p	DR(p)	p mod 6	DR(M_p)	In triad
20	4,423	4	1	1	no
21	9,689	5	5	4	yes
22	9,941	5	5	4	yes
23	11,213	8	5	4	yes
24	19,937	2	5	4	yes
25	21,701	2	5	4	yes
26	23,209	7	1	1	no
27	44,497	1	1	1	no
28	86,243	5	5	4	yes
29	110,503	1	1	1	no
30	132,049	1	1	1	no
31	216,091	1	1	1	no
32	756,839	2	5	4	yes
33	859,433	5	5	4	yes
34	1,257,787	1	1	1	no
35	1,398,269	2	5	4	yes
36	2,976,221	2	5	4	yes
37	3,021,377	5	5	4	yes
38	6,972,593	5	5	4	yes
39	13,466,917	1	1	1	no
40	20,996,011	1	1	1	no
41	24,036,583	4	1	1	no
42	25,964,951	5	5	4	yes
43	30,402,457	7	1	1	no
44	32,582,657	2	5	4	yes
45	37,156,667	5	5	4	yes
46	42,643,801	1	1	1	no
47	43,112,609	8	5	4	yes
48	57,885,161	5	5	4	yes
49	74,207,281	4	1	1	no
50	77,232,917	2	5	4	yes
51	82,589,933	2	5	4	yes
52	136,279,841	5	5	4	yes

Exponent list: the 52 Mersenne primes known as of the 2024 discovery of $M_{136279841}$ (GIMPS).

7. Mersenne numbers mod 999

2 has order 36 mod 999 (18 mod 27, 36 mod 37), so M_p mod 999 is fixed by p mod 36. Since 999 = 27 · 37 and $10^3 \equiv 1 \pmod{999}$, summing a number's comma groups gives its residue mod 999 directly; mod 27 and mod 37 follow from that one number. Example: M_{61} = 2,305,843,009,213,693,951; the groups sum to 3,016, and 3 + 016 = 19, matching the table ($61 \equiv 25 \pmod{36}$).

p mod 36	DR(p)	M_p mod 999	mod 27	mod 37	DR(M_p)	Known exponents
1	1	1	1	1	1	4
5	5	31	4	31	4	9
7	7	127	19	16	1	2
11	2	49	22	12	4	1
13	4	199	10	14	1	4
17	8	202	13	17	4	5
19	1	811	1	34	1	7
23	5	4	4	4	4	3
25	7	19	19	19	1	3
29	2	319	22	23	4	7
31	4	280	10	21	1	4
35	8	499	13	18	4	1
p = 2	2	3	3	3	3	1
p = 3	3	7	7	7	7	1

Every prime exponent above 3 falls in one of these 12 classes, so these 12 residues (plus 3 and 7) are the only values M_p can take mod 999. None is 0, so neither 27 nor 37 ever divides a Mersenne number with a prime exponent above 3. The mod-27 column repeats every 18 (rows 1 and 19 share residue 1, rows 5 and 23 share 4), while the mod-37 column distinguishes all 12 classes.